PAPER_381 — SGR1745 Compressed MUGE Spectral Term Decomposition & Perturbation Dominance Law

***Session:** 104 (Complete Re-Analysis — individual term values undiscovered in PAPER_372)*

PAPER_381 — SGR1745 Compressed MUGE Spectral Term Decomposition & Perturbation Dominance Law

Source: grok*share*11254865.txt, lines ~2900–2904 **Section:** SGR1745 compressed MUGE computation — full 8-term breakdown **Session:** 104 (Complete Re-Analysis — individual term values undiscovered in PAPER_372) **CP4 Class:** `SGR1745CompressedMUGESpectralTermDecompositionCalculator` (CP4 #32)

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1. Overview

PAPER_372 documented the final compressed MUGE result for SGR1745 (g ≈ 1.782×10^{3}■ m/s^2) and established the 8-function modular structure. However, it did NOT record the individual magnitudes of all 8 terms side-by-side. This paper fills that gap with the first complete **spectral term decomposition** showing the relative magnitude of each compressed MUGE contribution.

The key discovery: the **perturbation term dominates by 27 orders of magnitude** over the Newtonian base, revealing why the compressed model is **unphysical at magnetar scale**.

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2. SGR1745 System Parameters

Parameter	Symbol	Value	Units
Mass	M	2.984×10^{30}	kg
Radius	r	1×10■	m
Magnetic field	B	1×10^{10}	T
Critical B-field	B_crit	1×10^{11}	T
Age	t	3.799×10^{10}	s
Redshift	z	0.0009	—
Expansion velocity	v_exp	1×10^{3}	m/s
Dark matter mass	M_DM	1×10^{2}■	kg
Density contrast	δρ/ρ	0.1	—

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3. Complete 8-Term Spectral Decomposition

Term 1: Newtonian Base

```
g_\text{base} = \frac{GM}{r^2} = \frac{6.674\times10^{-11} \times
2.984\times10^{30}}{(10^4)^2}
```

```
\boxed{g_\text{base} = 1.991\times10^{12} \ \text{m/s}^2}
```

Term 2: Hubble Expansion Correction

```
g_\text{expansion} = g_\text{base} \times (1 + H_0 t)
```

At `t = 3.799\times10^{10}` s and `H_0 = 2.269\times10^{-18}` s^{-1}:

```
1 + H_0 t = 1 + (2.269\times10^{-18})(3.799\times10^{10}) = 1.0000000862
```

```
\boxed{g_\text{expansion\ factor} = 1.0000000862\ (\text{negligible at magnetar age})}
```

Term 3: Superconducting Correction (Meissner linear)

```
g_\text{SC} = g_\text{base} \times (1 + H_0 t) \times \left(1 -
\frac{B}{B_\text{crit}}\right)
```

```
\left(1 - \frac{10^{10}}{10^{11}}\right) = 0.9
```

```
\boxed{g_\text{SC\ adj} = 1.792\times10^{12} \ \text{m/s}^2}
```

Term 4: External BH Gravity (Ug3′)

```
U_{g3}' = \frac{GM_\text{BH}}{r_\text{BH}^2} \quad \text{(external Sgr A* at r = 26 kpc)}
```

```
\boxed{U_{g3}' = 6.746\times10^{-5} \ \text{m/s}^2}
```

Term 5: Cosmological Constant Floor

```
g_\Lambda = \frac{\Lambda c^2}{3} = \frac{1.1\times10^{-52} \times (3\times10^8)^2}{3}
```

```
\boxed{g_\Lambda = 3.3\times10^{-36} \ \text{m/s}^2 \quad (\text{effectively zero at
magnetar scale})}
```

Term 6: Quantum Coherence Term

```
g_\text{quantum} = \frac{\hbar}{\Delta x \cdot \Delta p} \cdot \int \psi^\dagger \hat{H}
\psi \, dV \cdot \frac{2\pi}{t_\text{Hubble}}
```

With `\Delta x \cdot \Delta p = 10^{-68}` $J^2 \cdot s^2$ and coherence integral = `2.176\times10^{-18}` J:

```
\boxed{g_\text{quantum} = 3.316\times10^{-35} \ \text{m/s}^2}
```

Term 7: Fluid Coupling

```
g_\text{fluid} = \rho_\text{fluid} \cdot V_\text{sys} \cdot g_\text{local}
```

```
\boxed{g_\text{fluid} = 4.189\times10^{-2} \ \text{m/s}^2}
```

Term 8: Dark Matter Perturbation (DOMINANT TERM)

```
g_\text{pert} = (M + M_\text{DM})\left(\frac{\delta\rho}{\rho} + \frac{3GM}{r^3}\right)
```

At `r = 10^4` m:

```
\frac{3GM}{r^3} = \frac{3 \times 6.674\times10^{-11} \times 2.984\times10^{30}}{(10^4)^3}
= 6.0\times10^{10}
```

```
(M + M_\text{DM}) \approx (2.984\times10^{30} + 10^{28}) \approx 3.0\times10^{30} \
\text{kg}
```

```
\boxed{g_\text{pert} = 1.782\times10^{39} \ \text{m/s}^2 \quad (\textbf{DOMINANT — 27
orders above base})}
```

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4. Complete Spectral Decomposition Table

Term	Formula	Value (m/s^2)	Orders above base
Base (Newtonian)	GM/r^2	1.991×10^{12}	—
SC adj (×0.9)	×(1−B/B_crit)	1.792×10^{12}	0
Ug3′ (ext. BH)	GM*BH*/*r*BH^2	$6.746\times10^{-}$■	−17
Cosmological floor	$\Lambda c^2/3$	3.3×10^{-3}■	−48
Quantum coherence	■■■■·2π/t_H	3.316×10^{-3}■	−47
Fluid coupling	ρ*f*·*V*·*g*loc	4.189×10^{-2}	−14
Perturbation (DM)	**(M+M_DM)(δρ/ρ+3GM/r^3)**	**1.782×10^{3}■**	**+27**

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5. Perturbation Dominance Law

Statement: For compact objects at `r \sim 10^4` m (magnetar scale), the dark matter perturbation term in the Compressed MUGE dominates by ≥ **27 orders of magnitude** over the Newtonian base.

Physical origin: The `3GM/r^3` factor scales as `r^{-3}` — making it catastrophically large at magnetar radii:

```
\frac{3GM}{r^3}\bigg|_{r=10^4} = 6.0\times10^{10} \gg \frac{3GM}{r^3}\bigg|_{r=1 \text{
AU}} \approx 1.7\times10^{-23}
```

Implication: The Compressed MUGE formulation is **unphysical** for compact objects. At magnetar scale, `r = 10^4` m violates the assumption that `(M+M_{DM})\cdot\frac{3GM}{r^3}` remains a small correction. The **Resonance MUGE model** (PAPER_371) with fluid-dominant term `a_{fluid\_freq} = 1.773\times10^{-9}` m/s^2 is the physically appropriate description.

Validity domain criterion:

```
\text{Compressed MUGE valid when: } \frac{3GM}{r^3} \ll \frac{\delta\rho}{\rho} \quad
\Rightarrow \quad r \gg \left(\frac{3GM}{\delta\rho/\rho}\right)^{1/3}
```

For SGR1745 with `\delta\rho/\rho = 0.1`:

```
r_\text{min\_compressed} = \left(\frac{3 \times 6.674\times10^{-11} \times
2.984\times10^{30}}{0.1}\right)^{1/3} \approx 1.3\times10^7 \ \text{m}
```

The magnetar radius `r = 10^4` m violates this by 3 orders — confirming Compressed MUGE is invalid.

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6. Connection to PAPER_379 Dual-Model Comparison

PAPER_379 showed the **48-order divergence** between compressed (`1.782\times10^{39}` m/s^2) and resonance (`1.773\times10^{-9}` m/s^2) results for SGR1745. This paper explains the **root cause**: the perturbation term alone produces a 27-order excess, and the two models diverge by 48 orders because compressed MUGE sums a physically explosive term that simply should not apply at `r = 10^4` m.

Model selection rule (first formal statement):

- `r < 10^7` m (compact stars, NS, BH): Use **Resonance MUGE**
- `r > 10^{10}` m (galaxies, clusters, cosmological): Use **Compressed MUGE**
- Transition zone: Both models compared, resonance preferred when fluid term dominates

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7. Key Equations

```
g_\text{compressed}^\text{SGR1745} = \underbrace{1.792\times10^{12}}_\text{Newt+SC} +
\underbrace{6.746\times10^{-5}}_\text{Ug3'} +
\underbrace{4.189\times10^{-2}}_\text{fluid} +
\underbrace{1.782\times10^{39}}_\text{perturbation}
```

```
\approx 1.782\times10^{39} \ \text{m/s}^2 \quad \text{(perturbation-dominated)}
```

Resonance MUGE (physically correct for this system):

```
g_\text{resonance}^\text{SGR1745} \approx a_{fluid\_freq} = 1.773\times10^{-9} \
\text{m/s}^2
```

Full divergence: `1.782\times10^{39} / 1.773\times10^{-9} \approx 10^{48}` — 48 orders of magnitude.

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8. References Within Codebase

- PAPER_372: Compressed UQFF B/Bcrit Superconductivity — framework and final value
- PAPER_371: MUGE 12-Term Resonance — correct model for SGR1745
- PAPER_379: Dual-model 7-system comparison — the 48-order divergence
- `CompressedUQFFBcritSuperconductivityCalculator` (CP4 #15): Full 8-function implementation

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